



Stress is associated with an increased risk of recurrent seizures in adults

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SUMMARY

Objective: The literature is sparse on the complex interrelationships between stressors, depression, anxiety disorders, and epilepsy. We hypothesized that a relationship exists between stress and epilepsy. We evaluated whether markers of stress are associated with seizure recurrence in a low income community-based cohort of adults with single unprovoked seizure or newly diagnosed epilepsy.

Methods: We ascertained adult residents of Northern Manhattan and Harlem, New York City, with a first unprovoked seizure or newly diagnosed epilepsy, between December 2010 and January 2013. At enrollment, we collected information about seizure phenomenology, demographics, clinical information, and measures of stress (environmental stress, stressful life events, facets of allostatic load—i.e., the cumulative effect of adaptation to stress, psychiatric disorders, and low collective efficacy). Collective efficacy assesses neighborhood characteristics and incorporates social cohesion and informal social control. All subjects were followed for 2 years for further seizures. Cox proportional hazard regression models were used to estimate the hazard ratios of seizure recurrence during the 2 years of follow-up.

Results: We identified 52 subjects (64.2%) with a single unprovoked seizure and 29 (35.9%) with newly diagnosed epilepsy. Seizure recurrence was recorded in 38.5% (N = 20) of subjects with a single unprovoked seizure and in 69% of those with epilepsy (N = 20) (p = 0.01). In the overall sample, the hazard of seizure recurrence was increased by lifetime generalized anxiety disorder (3.0-fold) and by low collective efficacy (2.7-fold). In a second model, the hazard was increased by lifetime mood disorder (2.1-fold) and low collective efficacy (2.5-fold).

Significance: Markers of stress (i.e., low collective efficacy, lifetime mood disorder, and lifetime generalized anxiety disorder) were associated with an increased risk for seizure recurrence in adults with a single unprovoked seizure or newly diagnosed epilepsy. Stress-reducing interventions, such as mindfulness, may be a useful, safe, and inexpensive adjunctive treatment for epilepsy.

KEY WORDS: Community-based, Epidemiology, Stress, Seizure recurrence.



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Stress is among the most frequent self-reported precipitating factors associated with seizures in patients with epilepsy.^{1–3} The stress response is complex and involves different systems and mediators, depending on the type and the duration of the stressor.^{4,5}

Stressful life events in the general population (e.g., divorce of parents, loss of a child, physical or emotional abuse by parents, loss of job) have been shown to predict both anxiety and depression.^{6,7} Neighborhood characteristics, as measured by collective efficacy,⁸ which incorporates

KEY POINTS

- Markers of stress are associated with seizure recurrence in adults with a single unprovoked seizure or incident epilepsy in a low-income community
- Stress management may be a useful adjunctive treatment for epilepsy accompanied by stress
- Well-designed and statistically powered studies are needed to determine the usefulness of behavioral interventions to reduce seizure recurrence

social cohesion (i.e., the willingness to intervene for the common good, based on mutual trust and solidarity) and informal social control (i.e., the capacity of a group to regulate its members through informal mechanisms to achieve public order), have been associated with depression.⁹ Stressful life events are also linked to incident epilepsy. For example, in a population-based study, the loss of a child was associated with an increased risk of developing epilepsy in the parents.¹⁰

History of depressive disorder has been shown to increase subsequent seizure frequency in epilepsy patients treated with antiepileptic drugs (AEDs).^{11,12} Studies have also shown that a lifetime history of psychiatric disorders is associated with continued seizures after mesial temporal lobectomy.^{13,14} However, the role of stress in these studies is unknown.

In addition, the literature is sparse on the complex interrelationships between stressors, mood disorders, anxiety disorders, and epilepsy.^{15,16} Allostatic load is the cumulative effect of adaptation to stress and has been studied in other chronic disease.¹⁷ The measure combines several parameters of cardiovascular function, levels of metabolism, levels of activity of the hypothalamic–pituitary–adrenal (HPA) axis, and indices of sympathetic nervous system activity (i.e., blood pressure, waist–hip ratio, serum cholesterol, glycosylated hemoglobin, dehydroepiandrosterone sulfate, cortisol, adrenalin, and noradrenalin).¹⁸ Social environment, indexed as low socioeconomic status, can influence the allostatic load of the individual.¹⁸ It is possible that stress may increase brain excitability through the effect of corticosteroid hormones, neurotransmitters, and neuropeptides such as corticotropin-releasing hormone (CRH).^{19,20} We hypothesized that a relationship exists between stress, environment, and recurrent seizures in epilepsy and after a first unprovoked seizure (Fig. 1).²¹

We evaluated whether stress, as measured by collective efficacy, stressful life events, facets of allostatic load, and psychiatric disorders, is associated with seizure recurrence in a cohort of adults with single unprovoked seizure or newly diagnosed epilepsy identified in the communities of Harlem and Northern Manhattan.

MATERIALS AND METHODS

Subjects

We ascertained residents of Northern Manhattan and Harlem, New York City, with a first unprovoked seizure or newly diagnosed epilepsy (zip codes 10026, 10027, 10030, 10031, 10035, 10037, and 10039) over a 2-year period (between 12/05/10 and 1/11/2013). Cases were ascertained from Columbia Presbyterian Medical Center and its community hospital, the Allen Pavilion, St. Lukes-Roosevelt Hospital Center, and Mount Sinai Medical Center. Other than private neurologists in these hospitals, there are no private neurologists with practices outside of the hospitals serving the Northern Manhattan and Harlem catchment area. Cases were drawn from emergency departments (EDs) at all hospitals, from Neurology and Epilepsy Clinics, and from electroencephalography (EEG) laboratories at Columbia's New York Presbyterian Hospital. Ascertainment methods were specific to each site, because there were differences in local practice.

Screening for potential incident subjects was done weekly by two research assistants. After review of medical records for potential incident subjects, those judged eligible received a letter signed by the ED director or their attending physician, explaining the study. After 2 weeks, if the potential subjects had not called to refuse, they were telephoned by a research assistant to confirm eligibility, using a semi-structured validated Seizure Screen,²² and to invite eligible individuals to participate. Subjects who agreed to participate were interviewed at enrollment²² about their seizure phenomenology, demographics, medical care use, comorbidities, quality of life, and stigma associated with epilepsy.

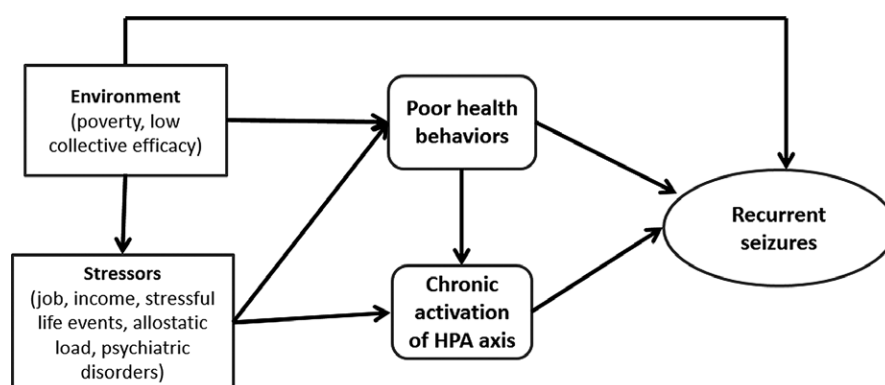
The assessment of seizure phenomenology was conducted by two trained research assistants (RAs) in person for 79 participants and over the phone for two participants. The Seizure Screen has sensitivity 96%, specificity 93%, and positive predictive value of 23%. All subjects were followed every 4 months for 2 years for further seizures, use of medical care, quality of life, stigma, and paid work. If the subject indicated they did not want to be contacted, could not be contacted after repeated attempts, or had died prior to the contact, then the medical record was abstracted for the seizure information. The RAs were not informed about the study hypotheses.

Each case was reviewed by three epileptologists (WAH, AP, and CH), who considered the medical record and an in-depth interview, and reached a consensus for the following: first unprovoked seizure or newly diagnosed epilepsy, seizure type, seizure syndrome, and seizure etiology.

Inclusion criteria

An unprovoked seizure was defined as a seizure or multiple seizures within a 24 h period without an identified proximate precipitant. Epilepsy was defined as recurrent unprovoked seizures.²³ Newly diagnosed subjects were followed for 2 years to assess seizure recurrence.

Figure 1.
Possible relationships between
environment, stressors, and
recurrent seizures.
Epilepsia © ILAE



Exclusion criteria

We excluded incident acute symptomatic seizures and incident or recurrent febrile seizures. Subjects residing in zip codes outside the community and children younger than 18 years of age were also excluded. Also excluded for this analysis were adult subjects who did not want to answer questions but had their medical records abstracted ($N = 35$) and those who refused to participate ($N = 25$).

This study was approved by the institutional review board at each institution. A certificate of confidentiality was issued by the National Institutes of Health. All participants in this analysis provided written informed consent.

Measures

Age

The age at diagnosis of a first unprovoked seizure or epilepsy was considered as a continuous variable.

Race

Race was categorized as Asian-Indian, Black, Mixed, Native American, White, and Other.

Ethnicity

Ethnicity was categorized as Hispanic or non-Hispanic.

Socioeconomic status

Socioeconomic status at enrollment (SES) was ascertained by income, health insurance, and employment. The community is predominantly of very low income, and annual household income was categorized as less than \$15,000, \$15,000–\$49,999, and \$50,000 and higher. Health insurance was categorized as state support, provided by employer, self-paid, or no coverage. These factors were separately used as indices of stress.

Markers of stress

Collective efficacy

We measured collective efficacy at enrollment, defined as social cohesion among neighbors combined with their willingness to intervene for the common good.⁸ When

measured on an individual level, collective efficacy is a measure of perceived environmental stress. Social control was measured using a 5-item Likert scale (1-very unlikely to 5-very likely) to determine how likely people perceive their community to intervene if (1) Children were skipping school, (2) Children were spray-painting graffiti on local buildings, (3) Children were showing disrespect to an adult, (4) A fight broke out in front of their house, and (5) The fire station closest to their house was threatened by budget cuts. An individual's perception about whether the community intervenes to suppress social disorder may add to information on the effect of SES on health outcomes in epilepsy. Social cohesion was measured with a 5-item Likert scale to determine how strongly people agree (1-strongly disagree to 5-strongly agree) that (1) People around here are willing to help their neighbors, (2) This is a close-knit neighborhood, (3) People in this neighborhood can be trusted, (4) People in this neighborhood generally don't get along with each other, and (5) People in this neighborhood do not share the same values. Responses to the two questionnaires, covering social control and social cohesion, were summed, and scores reversed for items 4 and 5 of social cohesion scale, to create a measure of collective efficacy.⁸ A cutoff of 34, the median value for all subjects, was used for the analysis; scores <34 indicated low levels of social control and social cohesion.

Stressful life events

Stressful life events were measured at enrollment through 13 questions drawn and adapted from the 55-item Stressful Life Events Inventory.²⁴ We selected the 13 items that were sources of major stress, including family illness and death, finances and job loss, relationships, physical injury, and legal matters. The presence of stressful life events was considered positive if the subject experienced at least one of the events in the previous 12 months.

Allostatic load

We measured three components of allostatic load¹⁷: (1) *Obesity (BMI)*. A body mass index (BMI) ≥ 30 was considered obesity. (2) *Obesity (waist-hip ratio)*. The ratio of waist circumference (measured at its narrowest point between the ribs and iliac crest) and hip circumference

(measured at the maximal buttocks) defined obesity when the ratio was >0.85 for women and >0.90 for men. These measurements were taken at enrollment. (3) *Blood pressure.* Systolic blood pressure (BP) and diastolic BP were measured at enrollment. High blood pressure was defined as systolic BP ≥ 140 mm Hg or diastolic BP ≥ 90 mm Hg.

Psychiatric disorders

The Structured Clinical Interview for DSM Disorders (SCID-I/P for DSM-IV [Diagnostic and Statistical Manual of Mental Disorders, Fourth Revision] Research Version, Patient version) was administered at enrollment by trained research assistants under the supervision of a psychiatrist and administered to adults at the enrollment interview to assess past and current history of mood disorders and generalized anxiety disorder.²⁵ These were then categorized as lifetime mood disorder and lifetime generalized anxiety disorder. Lifetime mood disorders included lifetime major depressive episode, dysthymia, bipolar disorder I, bipolar disorder II, manic episode, and hypomanic episode.

Use of antidepressant or anxiolytic drugs

At enrollment and follow-up, subjects were asked about the drugs they took, including psychotropic drugs. The presence of psychotropic drugs (including AEDs) was coded as yes or no.

Family history of depressive symptoms

We assessed history of depressive symptoms at enrollment in parents of subjects, using part of the Family History screen for depression.²⁶ We considered parents' depressive symptoms present if the subject reported that at least one of the parents ever had a period of at least 2 weeks feeling sad, blue, or depressed and/or feeling quite tired, having less energy, or not caring about usual activities.

Follow-up seizure interview

Over a 2-year follow-up period, subjects were interviewed over the phone every 4 months about any new seizures and their medical care for seizures.

A recurrent seizure was defined as the first unprovoked seizure occurring >24 h after the first unprovoked seizure or epilepsy diagnosis.

Statistical analyses

Descriptive statistical analyses were performed to analyze demographic and clinical characteristics, using frequencies and percentages to summarize categorical variables and median and interquartile range (IQR) to summarize continuous variables. Statistical significance was determined using the chi-square test or Fisher's exact test for categorical variables and Wilcoxon rank-sum test for continuous variables. The 0.05 level of significance and two-sided tests were used for all analyses.

Cox proportional hazard regression models were used to estimate the hazard ratios (HRs) of seizure recurrence during the 2 years of follow-up, among the whole sample and separately for single unprovoked seizure or epilepsy. Subjects were censored at loss to follow-up or at 24 months if there was no recurrence. Each factor was examined in univariate Cox models. Multivariable Cox models for all subjects were adjusted for age at the index date, gender, and epilepsy status; other models were adjusted only for age and gender when epilepsy and single unprovoked seizure were separately examined. The variables included in the models were those clinically relevant and significantly associated ($p < 0.05$) in the univariate analysis.

Two final models were created. One examined lifetime generalized anxiety disorder and level of collective efficacy as risk factors for seizure recurrence. A second model examined lifetime mood disorder and level of collective efficacy. Subjects were censored at their last date of follow-up, death, or the end of the study period, whichever came first.

The proportional hazard assumption was tested graphically and by testing for interaction.

All analyses were conducted using SAS software, v 9.4 (SAS Institute, Cary, NC, U.S.A.).

RESULTS

Among the 225 subjects identified, 136 agreed to be interviewed and 85 of these were adults. Four subjects were excluded because they did not have a first single unprovoked seizure or newly diagnosed epilepsy.

First unprovoked seizures and newly diagnosed epilepsy were therefore identified in 81 adult residents of Northern Manhattan and Harlem during the study period. Of these subjects, 52 (64.2%) had a single unprovoked seizure and 29 (35.9%) were newly diagnosed with epilepsy (Table 1). Among subjects with a single unprovoked seizure 20 (38.5%) had a second seizure during the follow-up, whereas among those with newly diagnosed epilepsy 20 (69%) had a seizure recurrence ($p = 0.01$).

There was no difference in demographic factors between interviewed subjects and those who either refused to participate in the interview or who did not respond to phone calls. Medical records were abstracted for these latter subjects.

There was no difference between subjects with a single unprovoked seizure and those with epilepsy for any of the other demographic, clinical, and psychiatric factors considered. However, the Elixhauser Index showed a difference between 54 subjects <65 years (median = 0; IQR 0–1 years) versus 23 subjects ≥ 65 years (median = 6; IQR 0–7 years; $p = 0.004$).

Overall, the median age at diagnosis was 51.4 years (IQR 29.2–70.9 years). There was no difference in median age between subjects with single unprovoked seizure and those

Table 1. Demographics, epilepsy-related characteristics, and psychiatric disorders in adults with single unprovoked seizure or newly diagnosed epilepsy

	Single unprovoked seizure N = 52 N (%)	Newly diagnosed epilepsy N = 29 N (%)	p-Value	Single unprovoked seizure or newly diagnosed epilepsy N = 81 N (%)
Gender				
Male	28 (53.9)	13 (44.8)	0.5	41 (50.6)
Female	24 (46.1)	16 (55.2)		40 (49.4)
Age at index date				
Median (IQR)	55.3 (28.2–75.8)	49.8 (32.3–62.7)	0.6	51.4 (29.2–70.9)
Ethnicity				
Hispanic	32 (61.5)	22 (75.9)	0.2	54 (66.7)
Non-Hispanic	20 (38.5)	7 (24.1)		27 (33.3)
Race				
Asian Indian	1 (1.9)	0 (0)	0.5	1 (1.2)
Black	12 (23.1)	7 (24.1)		19 (23.5)
Mixed	4 (7.7)	5 (17.2)		9 (11.1)
Native American	1 (1.9)	0 (0)		1 (1.2)
Other	18 (34.6)	12 (41.5)		30 (37.1)
White	16 (30.8)	5 (17.2)		21 (25.9)
Annual household income ^a				
<\$15,000	17 (32.7)	9 (32.1)	0.91	26 (32.5)
\$15,000–49,999	31 (59.6)	16 (57.1)		47 (58.8)
≥\$50,000	4 (7.7)	3 (10.7)		7 (8.7)
Employment				
Yes	25 (48.1)	13 (44.8)	0.7	38 (46.9)
No	27 (51.9)	16 (55.2)		43 (53.1)
Health insurance				
State including Medicaid	39 (75.0)	22 (75.9)	0.9	61 (75.3)
Employer	7 (13.5)	5 (17.2)		12 (14.8)
Self-paid	2 (3.8)	0 (0)		2 (2.5)
No coverage	4 (7.7)	3 (6.9)		6 (7.4)
Time to recurrence (months)				
median (IQR)	12.2 (3.7–12.2)	2.3 (0.6–12.2)	0.003	8.1 (1.4–12.2)
Seizure recurrence				
Yes	20 (38.5)	20 (69.0)	0.01	40 (49.4)
No	32 (61.5)	9 (31.0)		41 (50.6)
Allostatic load				
Obesity (BMI) ^b				
Yes	13 (27.1)	8 (32.0)	0.72	21 (28.8)
No	35 (72.9)	17 (68.0)		52 (71.2)
Obesity (waist–hip) ^c				
Yes	27 (64.3)	18 (72.0)	0.5	45 (67.2)
No	15 (35.7)	7 (28.0)		22 (32.8)
Hypertension ^d				
Yes	20 (40.8)	13 (44.8)	0.7	33 (42.3)
No	29 (59.2)	16 (55.2)		45 (57.7)
Lifetime major depressive episode				
Yes	11 (21.1)	9 (31.0)	0.3	20 (24.7)
No	41 (78.9)	20 (69.0)		61 (75.3)
Lifetime generalized anxiety disorder				
Yes	6 (11.5)	2 (6.9)	0.7	8 (9.9)
No	46 (88.5)	27 (93.1)		73 (90.1)
Lifetime mood disorders				
Yes	17 (32.7)	9 (31.0)	1.0	26 (32.1)
No	35 (67.3)	20 (69.0)		55 (67.9)
History of depressive symptoms in parents ^e				
Yes	18 (41.9)	11 (50.0)	0.63	29 (44.6)
No	25 (58.1)	11 (50.0)		36 (55.4)

Continued

Table 1. Continued.

	Single unprovoked seizure N = 52 N (%)	Newly diagnosed epilepsy N = 29 N (%)	p-Value	Single unprovoked seizure or newly diagnosed epilepsy N = 81 N (%)
Collective efficacy ^f				
Low level (<34)	25 (48.1)	14 (48.3)		39 (48.1)
High level	27 (51.9)	15 (51.7)	1.0	42 (51.9)
Any stressful life events				
Yes	38 (73.1)	22 (75.9)	0.9	60 (74.1)
No	14 (26.9)	7 (24.1)		21 (25.9)
Elixhauser Index				
Median (IQR)	0 (0–5)	0 (0–7)	1.0	0 (0–5)

^a1 missing; ^b8 missing; ^c14 missing; ^d3 missing; ^e16 missing.

^fLevels of social cohesion and social control.

with epilepsy (Table 1). Forty-one subjects (50.6%) were male.

The median time between index date and the enrollment assessment was 63 days (IQR: 38.5–99.5 days).

Markers of Stress

Demographics (Table 1)

Among interviewed subjects, 32.5% (N = 26) were living in households with an annual income less than \$15,000 and 58.8% (N = 47) with an annual income between \$15,000 and \$49,000. Most subjects had health insurance provided by the state (75.3%), and 46.9% were employed. The study group is mostly Hispanic (66.7%).

Body measurements

Among the subjects, 21 (28.8%) were obese as assessed by the BMI. Using the waist–hip ratio, 45 (67.2%) were obese. Hypertension was diagnosed in 42.3%.

Psychiatric disorders

Twenty subjects (24.7%) met DSM-IV criteria for lifetime major depressive episode, eight (9.9%) subjects had a diagnosis of lifetime generalized anxiety disorder, and 26 (32.1%) of lifetime mood disorders (Table 1).

Collective efficacy

The level of collective efficacy was low for 39 subjects (48.1%). There was no association between collective efficacy and annual household income.

Life stress scale

Sixty subjects (74.1%) experienced at least one stressful life event (Table 1).

Risk of recurrent seizures

Among subjects with a single unprovoked seizure, the median time to recurrence was 12.2 months (IQR: 3.7–

12.2) and for those with epilepsy it was 2.3 months (0.6–12.2) (p = 0.003) (Table 1).

Univariate models (Table 2)

No factor was associated with an increased hazard ratio for seizure recurrence in subjects with single unprovoked seizure and those with newly diagnosed epilepsy. However, adjusting for age at diagnosis, gender, and diagnosis of epilepsy versus single unprovoked seizure, the hazard for seizure recurrence in the overall sample was increased by low levels of collective efficacy, lifetime major depression, and lifetime generalized anxiety disorder (Table 2).

Multivariable models (Table 3)

Lifetime generalized anxiety disorder, lifetime mood disorder, and collective efficacy were associated with an increased hazard of seizure recurrence in the whole population, adjusting for age at the index date, gender, and epilepsy versus single unprovoked seizure. In the overall sample, the hazard of seizure recurrence was significantly increased for collective efficacy and lifetime generalized anxiety disorder or lifetime mood disorder. The hazard of seizure recurrence was significantly increased for both models.

Both models were also evaluated separately in subjects with a single unprovoked seizure and newly diagnosed epilepsy. Estimates were similar but did not reach statistical significance except for the model including lifetime generalized anxiety disorder in subjects with epilepsy (Table 3).

DISCUSSION

This is the first study to follow a representative cohort of adults with single unprovoked seizure or newly diagnosed epilepsy in a defined community to assess whether markers of stress are associated with seizure recurrence. We found that low collective efficacy, and both lifetime mood disorders and lifetime generalized anxiety disorder were associated with seizure recurrence in adults with single

Table 2. Univariate Cox models for risk of recurrent seizure in single unprovoked seizure and newly diagnosed epilepsy

	Single unprovoked seizure ^a HR (95% CI), p-value	Newly diagnosed epilepsy ^a HR (95% CI), p-value	Single unprovoked seizures and newly diagnosed epilepsy ^b HR (95% CI), p-value
Blood pressure			
Hypertensive	0.6 (0.2–1.8), p = 0.4	1.5 (0.5–4.2), p = 0.4	1.0 (0.5–2.0), p = 1.0
Not hypertensive (ref)	1.0	1.0	1.0
BMI			
Obese	2.2 (0.9–5.6), p = 0.1	1.0 (0.4–2.5), p = 1.0	1.6 (0.8–3.0), p = 0.2
Not obese (ref)	1.0	1.0	1.0
Collective efficacy			
Low level ^{dc}	2.7 (0.9–7.8), p = 0.06	2.3 (0.9–5.6), p = 0.07	2.6 (1.3–5.0), p = 0.004
High level (ref)	1.0	1.0	1.0
Any stressful life event			
Yes	1.6 (0.4–5.8), p = 0.5	1.7 (0.6–5.2), p = 0.3	1.7 (0.8–4.0), p = 0.2
No (ref)	1.0	1.0	1.0
Lifetime major depression episode			
Yes	1.6 (0.6–4.4), p = 0.3	2.3 (0.9–5.8), p = 0.09	2.1 (1.1–4.1), p = 0.03
No (ref)	1.0	1.0	1.0
Lifetime generalized anxiety disorder			
Yes	2.5 (0.8–7.7), p = 0.1	3.8 (0.8–18.3), p = 0.1	2.7 (1.1–6.7), p = 0.03
No (ref)	1.0	1.0	1.0
Lifetime mood disorders			
Yes	2.0 (0.8–4.8), p = 0.1	2.3 (0.9–5.8), p = 0.09	2.3 (1.2–4.3), p = 0.01
No (ref)	1.0	1.0	1.0
History of depressive symptoms in parents			
Yes	1.7 (0.6–4.6), p = 0.3	0.10 (0.3–2.6), p = 0.8	1.3 (0.6–2.6), p = 0.5
No (ref)	1.0	1.0	1.0
Elixhauser index	1.1 (0.9–1.2), p = 0.4	0.98 (0.9–1.1), p = 0.6	0.99 (0.9–1.1), p = 0.8

^aAdjusted for age and gender; ^badjusted for age, gender, epilepsy versus single unprovoked seizure; ^cbelow the median of 34; ^dlevels of social cohesion and social control.

Table 3. Multivariable Cox regression for mood disorders, generalized anxiety disorder, and collective efficacy

	Single unprovoked seizure ^a Adjusted HR (95% CI)	Epilepsy ^a Adjusted HR (95% CI)	All adults ^c Adjusted HR (95% CI)
Model 1			
Lifetime generalized anxiety disorder			
Yes	2.4 (0.8–7.7), p = 0.1	5.3 (1.01–27.6), p = 0.049	3.0 (1.2–7.6), p = 0.02
No (ref)	1.0	1.0	1.0
Collective efficacy			
Low level ^{db}	2.6 (0.9–7.4), p = 0.07	2.6 (1.0–6.6), p = 0.04	2.7 (1.4–5.2), p = 0.003
High level (ref)	1.0	1.0	1.0
Model 2			
Lifetime mood disorder			
Yes	1.9 (0.8–4.7), p = 0.2	2.2 (0.8–5.7), p = 0.1	2.1 (1.1–4.0), p = 0.02
No (ref)	1.0	1.0	1.0
Collective efficacy			
Low level ^{db}	2.7 (0.9–7.5), p = 0.07	2.2 (0.9–5.6), p = 0.08	2.5 (1.3–4.8), p = 0.007
High level (ref)	1.0	1.0	1.0

^aAdjusted for age at diagnosis, gender; ^bbelow the median of 34; ^cadjusted for age at diagnosis, gender, and epilepsy; ^dof social cohesion and social control.

unprovoked seizure or incident epilepsy. Obesity and hypertension were not associated with seizure recurrence.

Collective efficacy, a measure of perceived environmental stress considered at an individual level,²⁷ has not been

previously studied in epilepsy. However, other studies have shown that patients with epilepsy consider stress as a common trigger for seizure recurrence.^{16,28} So too, the effect of environmental stressors (i.e., evacuation due to flood, 9/11

terror attack) on seizure recurrence in prevalent epilepsy shows an increase in seizure frequency among exposed subjects.^{2,3} A study of adults with prevalent epilepsy did not find an association between stressful life events and seizure frequency.²⁹ Our study differs from these as it considers a community-based study of incident single unprovoked seizure and incident epilepsy rather than investigations of prevalent epilepsy. Despite this difference, both incident and prevalent populations show an increased seizure recurrence in the presence of perceived stress.

Stress responses are complex and involve different systems and mediators depending on the type and the duration of the stressor,^{4,5} and chronic stress has been shown to determine changes in brain structure and function.^{30–32} Animal models of seizures have shown that early postnatal stress increases seizure susceptibility.^{33,34} One recent study demonstrated that chronic early life stress, provokes hyperexcitability (EEG spikes, seizures) in developing rats.²⁰

One possible explanation for our findings is that changes in neurotransmitter and hormone levels within the brain could influence the course of epilepsy, through the effect on synapses that may alter neuronal excitability and promote seizures.^{28,35,36} Others have shown that activation of the HPA axis associated with stress is associated with an increase in epileptiform activity in animals with preexisting epilepsy.^{35,37} However, mechanisms are unknown in people with epilepsy.

Our study findings are parallel with those of a large clinical series of newly diagnosed and treated people with epilepsy in which a lifetime history of depression was associated with an increased risk for intractable epilepsy.¹² Similarly another prospective study of consecutive patients with new-onset epilepsy showed that those subjects who endorsed more symptoms of depression and anxiety before starting an AED were less likely to be pharmacoresponsive after 12 months of follow-up and had a recurrent seizure earlier than those with fewer symptoms.¹¹ In some studies, a lifetime history of depression is also associated with continued seizures, even after surgical removal of the epileptogenic anterior temporal lobe,^{13,14} although other studies did not find the same association.^{38,39} In another surgical study, preoperative anxiety disorder was associated with persistent simple partial seizures.³⁹

If stress is associated with seizure recurrence, then studies to decrease perceived stress in people with epilepsy may show a lower seizure recurrence. In this regard, stress management through behavioral intervention may be a useful adjunctive treatment for epilepsy accompanied by stress. In the few studies that have been done, sample sizes are small, duration of treatment is variable, and rarely are patients blinded. These studies have looked at the effects of psychological interventions such as relaxation therapy, cognitive behavioral therapy, biofeedback, and education as additional treatment in epilepsy. In some studies, there was a statistically significant difference in

seizure numbers between groups.⁴⁰ However, a Cochrane review was unable to establish a clear effect of such interventions on seizure recurrence.⁴¹

Strengths and limitations

Strengths: Ours is a prospective community-based study of incident cases of single unprovoked seizure or newly diagnosed epilepsy. A strength of our study was that recall bias was diminished through the collection of risk factor information at the time of enrollment with the exception of lifetime mood disorder and lifetime generalized anxiety disorder. Moreover, all psychiatric diagnoses were obtained using a standardized structured interview (SCID-IV).

Limitations: Follow-up information on seizure recurrence and medication adherence was collected based on self-report. Other studies have found that seizures are underreported if they are subtle,⁴² as the patient may be less aware of them. Subjects with auras or with absence seizures may therefore underreport seizures. We considered the first seizure recurrence in this study. However, the time to reported recurrence might have been longer if a fraction of the subjects did not know if they had a recurrent seizure.

It was not possible to collect all facets of allostatic load because of insufficient funding. Thus only information on obesity, blood pressure, and waist–hip ratio was available to analyze such items.

We had no access to information on counseling or psychological treatments. Therefore, we were unable to include this information as a possible confounder. Another possible limitation is the wide age range of our sample. All markers of stress considered and medical and psychiatric comorbidities assessed could have different impacts on seizure recurrence, depending on the age group. However, given the small sample we were unable to analyze each age group separately.

The Harlem and Northern Manhattan community is characterized by low SES, a high prevalence of Hispanics, and a small range of variability in demographics. This homogeneity limits the broader generalizability of our findings. Although our sample size was not large, we were able to detect differences in recurrence risk. About 40% of identified subjects refused to be interviewed; however, their demographics did not differ from those who participated in the study interviews. Therefore, the subjects analyzed were an unselected group of the full population.

CONCLUSION

Stress-reducing interventions for seizure recurrence, such as relaxation therapy, mindfulness, cognitive behavioral therapy, and biofeedback, may be considered as adjunctive treatments that are safe and inexpensive. This is useful for those with low incomes, as relaxation and mindfulness have shown efficacy in other patient populations.⁴³ There

remains a need for well-designed and statistically powered studies to determine whether behavioral intervention reduces stress and seizure recurrence.

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DISCLOSURE OF CONFLICT OF INTEREST

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